Proposal Defense

Socratic AI: Agentic AI for Fostering Critical Thinking Education

A multi-agent tutor that asks instead of answers, and measures how well the learner reasons

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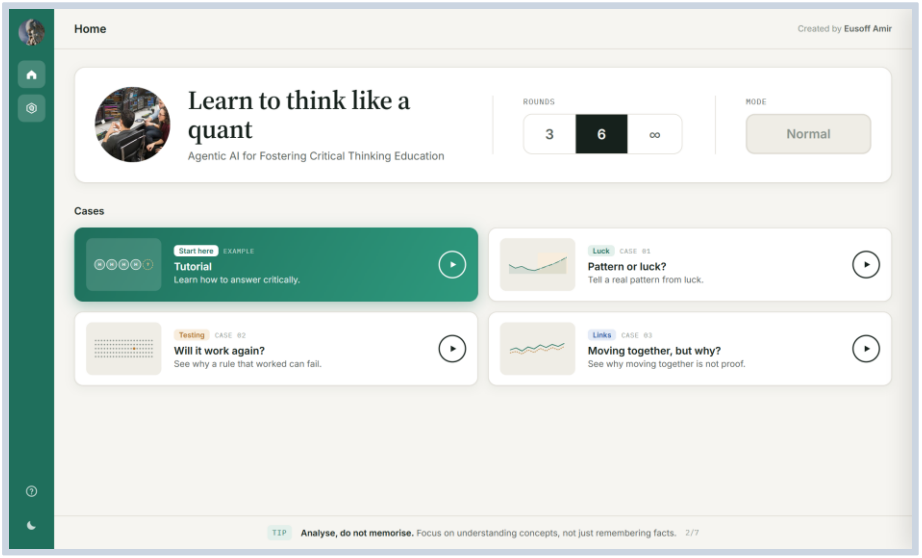
Course : EFB4042 Final Year Project I

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OUTLINE OF PRESENTATION

1. Introduction
2. Problem Statement, Objectives, Scope of Study
3. Literature Review
4. Proposed Methodology
5. Gantt Chart & Milestones
6. Preliminary Results
7. Conclusion



The working prototype, live online

INTRODUCTION

Generative AI now gives students a finished answer in seconds, so they can skip the thinking.

62%

of students use generative AI at least weekly [12]

319

knowledge workers studied: more trust in AI meant less critical thinking [10]

4

costs of "outsourced reasoning": fewer ideas, less effort, weaker memory, less reflection [11]

The idea of this project

Turn the same AI around. Instead of an answer machine, build a tutor that asks Socratic questions in a set order, so the learner builds the understanding, and then measure the quality of the learner's reasoning against a framework.

PROBLEM STATEMENT

Two Gemini Gem tutors were tested (an online Socratic Tutor and the first project prompt). Three problems were found.

1

No structure

Questions do not climb step by step from clarifying to questioning the question itself.

2

No assessment

The tutor cannot say how well the learner reasoned against any framework.

3

Poor experience

It repeated questions, forgot answers, ignored frustration and never said when it would end.

Real tester messages: *"bro i explained this before"* and *"you are asking way too much follow up questions"*

So a single Socratic prompt is not enough. The tutor needs a structured ladder, a separate assessor, memory, and a clear goal, progress and end.

OBJECTIVES

1. To design and build Socratic AI, a multi-agent tutoring prototype that teaches beginners one quant topic through Socratic dialogue.
2. To guide the tutor through six levels of Socratic questioning, and to assess the learner with the nine intellectual standards.
3. To show which type of reasoning (inductive, deductive, abductive) and which types of knowledge the learner used, measured over many questions.
4. To adapt the difficulty and support to each learner, with a clear goal, feedback after each answer and a progress view.
5. To evaluate the prototype against the baseline tutors from early testing.

SCOPE OF STUDY

1

One quant topic

Telling a real pattern in data from luck, for beginners, with no complex machine learning. Three short cases.

2

Multi-agent tutor

Four AI agents (Assessor, Planner, Tutor, Checker) guarded by plain-code rules, on the free Gemini model.

3

Framework assessment

6 Socratic levels and 9 standards now, then reasoning type and knowledge type, with a learner dashboard.

Target learners: university students new to quant (early testers are Dr Ho's students).

Future work, outside FYP: a second topic, real reading material with retrieval (RAG) [15], knowledge graphs, and a full class study (needs a paid key and ethics approval).

LITERATURE REVIEW

Agentic AI Building Blocks

Technique	Year	Core idea	Use in this project	Ref
Prompting	2025	A prompt is an instruction plus context; output depends on how the context is built	Every agent gets a structured JSON prompt	[16]
Chain of Thought	2022	Break a task into intermediate reasoning steps	The Planner reasons before choosing the next move	[17]
ReAct	2023	Reason, act, observe, then repeat	The Tutor runs a simulation, reads the result, then replies	[18]
RAG	2023	Ground answers in retrieved knowledge (naive, advanced, modular)	Case notes now; real RAG and knowledge graphs are future work	[15], [19]
Model Context Protocol	2024	A standard way to give an LLM tools, resources and prompts	Future: query a knowledge graph and learner records	[20]
Multi-agent systems	2024	Specialised agents collaborate (MetaGPT, AgileCoder)	Assessor, Planner, Tutor and Checker roles	[21], [22]

Based on the supervisor's course slides [16], [23] and the DeepLearning.AI short courses he recommended [24].

LITERATURE REVIEW

Frameworks for Critical Thinking

Framework	Year	Core idea	Role in this project	Limitation	Ref
Paul's 6 Socratic question levels	-	Clarify, assumptions, evidence, viewpoints, implications, question the question	The ladder: what the Tutor asks next	Says what to ask, not how good the answer is	[1]
Paul-Elder 9 intellectual standards	-	Clarity, accuracy, precision, relevance, depth, breadth, logic, significance, fairness	The Assessor marks each as 1 or 0	Does not say which type of reasoning was used	[1]
Paul-Elder 8 elements of thought	-	Purpose, question, information, inference, concepts, assumptions, implications, point of view	Prerequisites before reasoning (planned)	Links to reasoning types still open	[1]
Revised Bloom's taxonomy	2001	Knowledge dimension: factual, conceptual, procedural, metacognitive	Types of knowledge used (planned)	No causal or relational type	[2]
SOLO taxonomy	1982	Prestructural to extended abstract outcomes	Complexity: how many knowledge types are joined	Needs adapting to dialogue	[3]

LITERATURE REVIEW

Socratic Questioning and Socratic AI

Study	Year	Setting	Main finding	Limitation	Ref
Sahamid	2016	Classroom action research	Socratic questioning built critical thinking	Small, one class	[4]
Makhene	2019	Nursing education	Socratic inquiry helps critical thinking	Needs skilled teachers' time	[5]
Ho, Chen and Li	2023	Healthcare students	Socratic method improved thinking skills	Human tutor, does not scale	[6]
Degen and Asanov	2025	65 pre-service teachers, Socratic AI vs plain chatbot	More support for critical, reflective thinking; argues for multi-agent systems	Self-reported, not a direct measure	[7]
Fan	2025	Socratic scaffolding through AI	Structured scaffolding deepened inquiry	No link to reasoning constructs	[8]
Lee, Ang and Lai	2025	Bloom labels on chatbot dialogues	Dialogue logs reveal thinking processes	Descriptive only	[9]
Lee et al. (CHI)	2025	319 knowledge workers	More AI trust, less critical thinking	Survey, self-reported	[10]

LITERATURE REVIEW

Existing Socratic AI Tutors

Tool	Refuses answers	Question ladder	Framework assessment	Goal, progress, end	Reasoning and knowledge types
Online Gemini Gem "Socratic Tutor" (Test #1)	Yes	No	No (informal points)	No	No
Project prompt v1 (Test #2)	Yes	Partly	Partly (hidden 1 to 3 score)	No	No
Khanmigo, custom GPTs *	Yes	Not shown	Not shown	Partly	No
Socratic AI (this project)	**Yes**	**Yes, 6 levels**	**Yes, 9 standards**	**Yes**	**Planned**

** From public descriptions, not tested in this project.*

No tested tool combines a question ladder, framework assessment and a clear session, and none shows which type of reasoning a learner used.

LITERATURE REVIEW

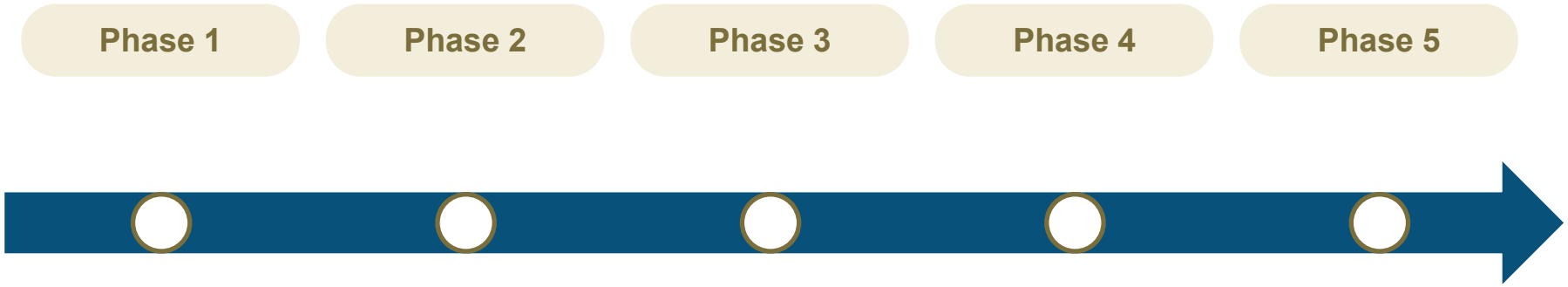
RESEARCH GAP

- AI use can reduce critical thinking effort, and current tutors are mostly answer machines.
- Socratic AI tutors are promising, but their questions have no principled ladder.
- Reasoning is rarely measured against a defined framework, and evidence is mostly self-reported.
- No tool shows the learner which type of reasoning or knowledge they used.

JUSTIFICATION

- A multi-agent tutor where the Tutor follows Paul's six levels, so questions climb in a set order.
- A separate Assessor marks every answer against the nine standards, with evidence, so it can be checked against human markers.
- Plain-code rules guard every AI decision, so steps are fair and testable ("agents decide, rules guard").
- Next, new agents add reasoning type and knowledge type, measured over many questions.

PROPOSED METHODOLOGY



Early testing
Two Gemini Gem tutors tested; problems written down

Requirements
9 design requirements R1 to R9 from the tests

Build
4 agents, rules, 3 cases, simulations, web app

Test and fix
Real sessions read turn by turn; problems fixed

Evaluate
Requirements, Assessor vs humans, learners

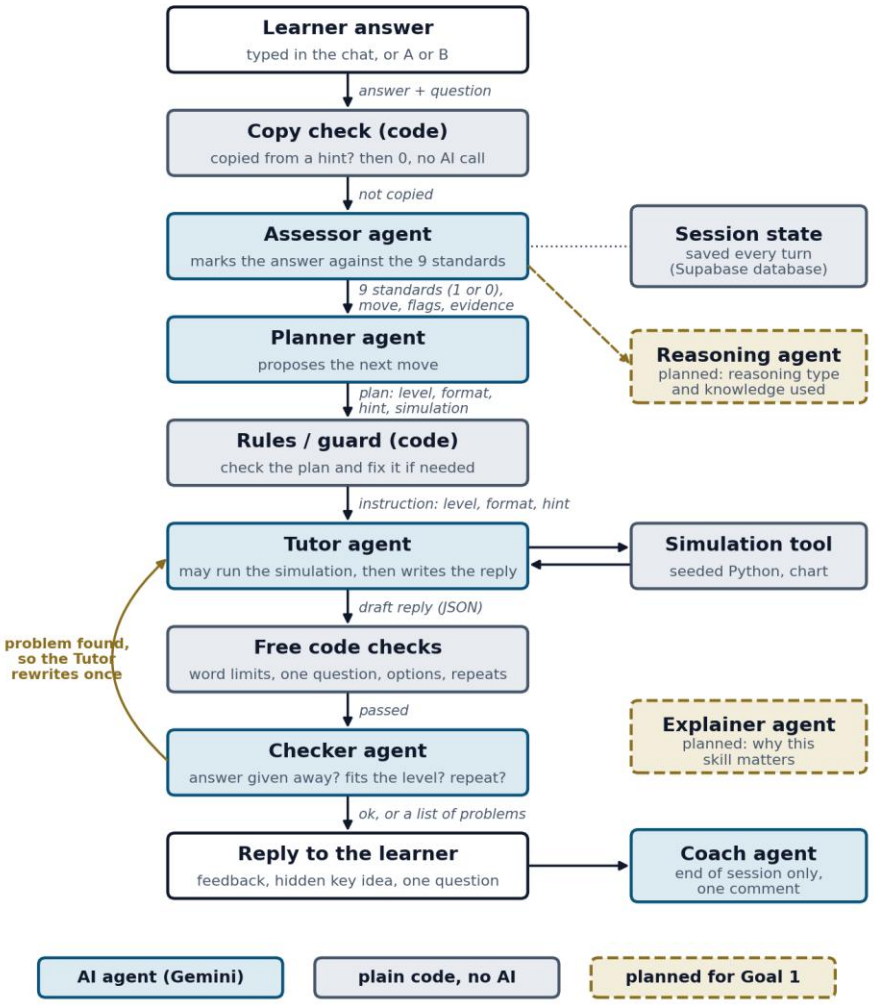
R1 Clear mission	R2 Fixed rounds, progress	R3 Remember, never re-ask
R4 Feedback and teaching	R5 Notice frustration	R6 Vary formats
R7 Separate assessor	R8 Learner stats	R9 Stable topic

Software
- Python 3.13, FastAPI, LangGraph
- Gemini 3.5 Flash Lite (free tier)
- HTML and JavaScript front end
- Supabase database, Render hosting (free)
- pytest: 94 automatic tests

Hardware
- A normal laptop; the AI runs in the cloud, so no GPU is needed

PROPOSED METHODOLOGY

System Architecture: Agents Decide, Rules Guard

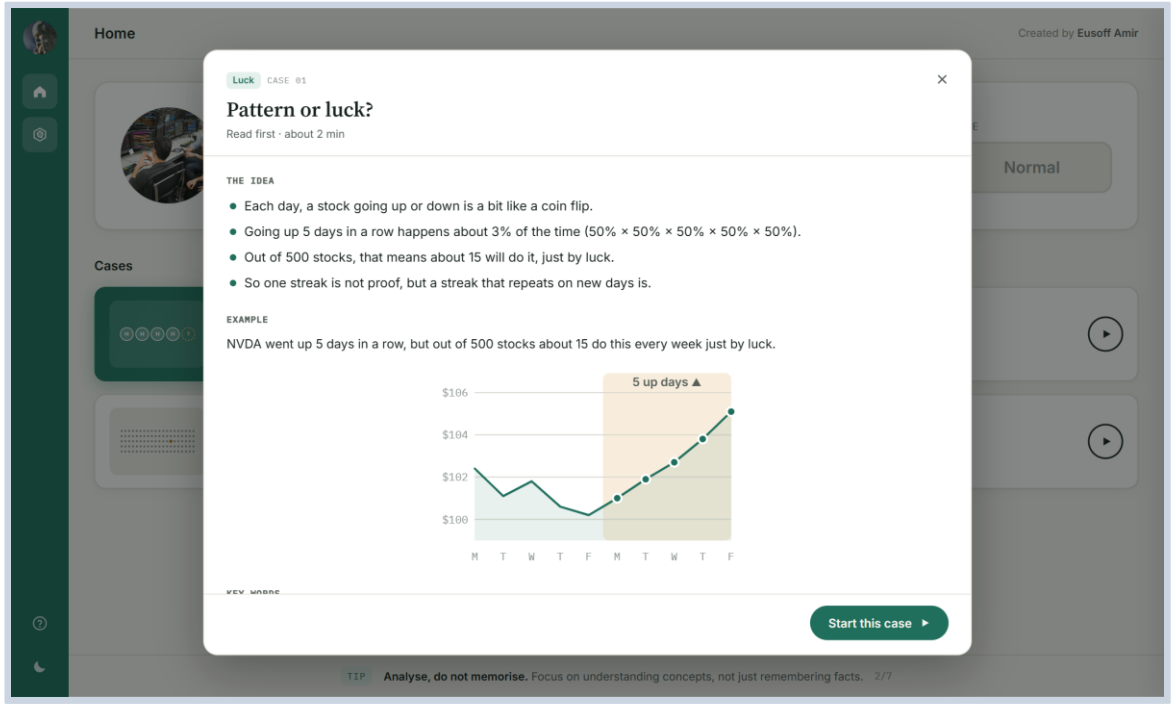


1. **Assessor:** marks each answer, 9 standards as 1 or 0, with an evidence quote
2. **Planner:** proposes the next level, question format, hint and simulation
3. **Rules (code):** a frustrated learner always gets an easy A or B question; no level up after a weak answer
4. **Tutor:** short feedback, a hidden hint, one question; may run a quant simulation
5. **Checker:** stops replies that give away the answer or repeat a question
6. **Coach:** one comment at the end on how the learner answered

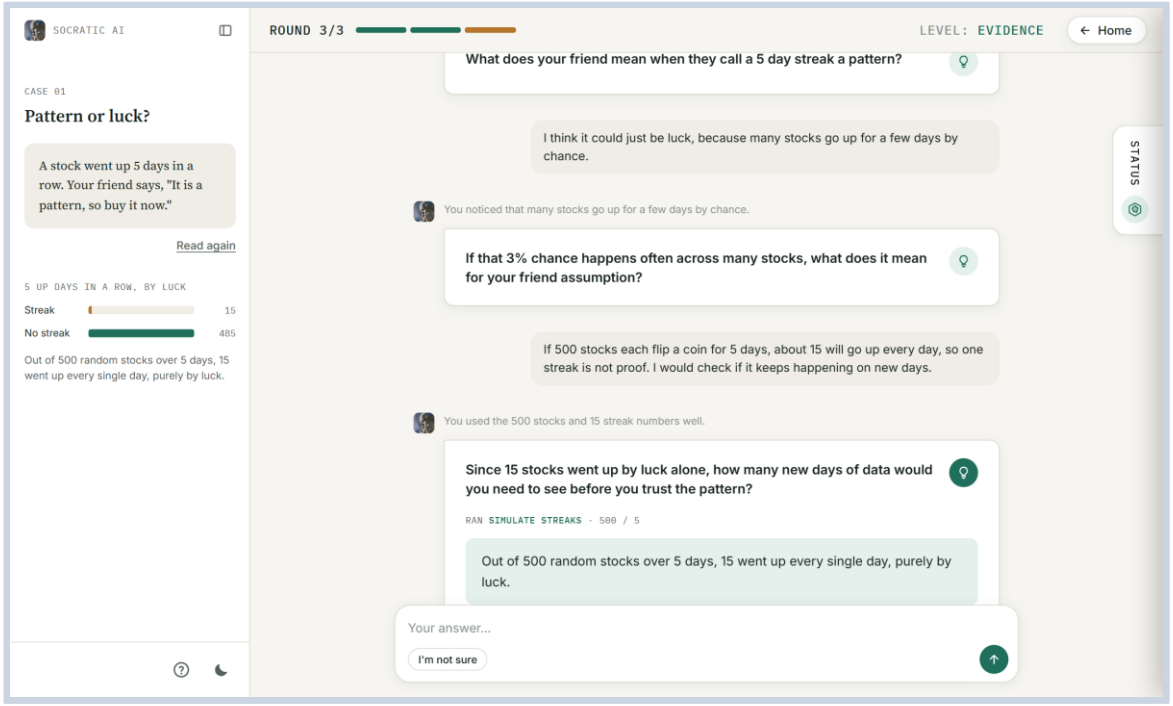
Each agent gets a small JSON payload built by code and must reply in a fixed JSON shape. About 25 AI calls per 6-round session.

PROPOSED METHODOLOGY

Session Design (R1 to R9)



1. A short reading and a mission before each case

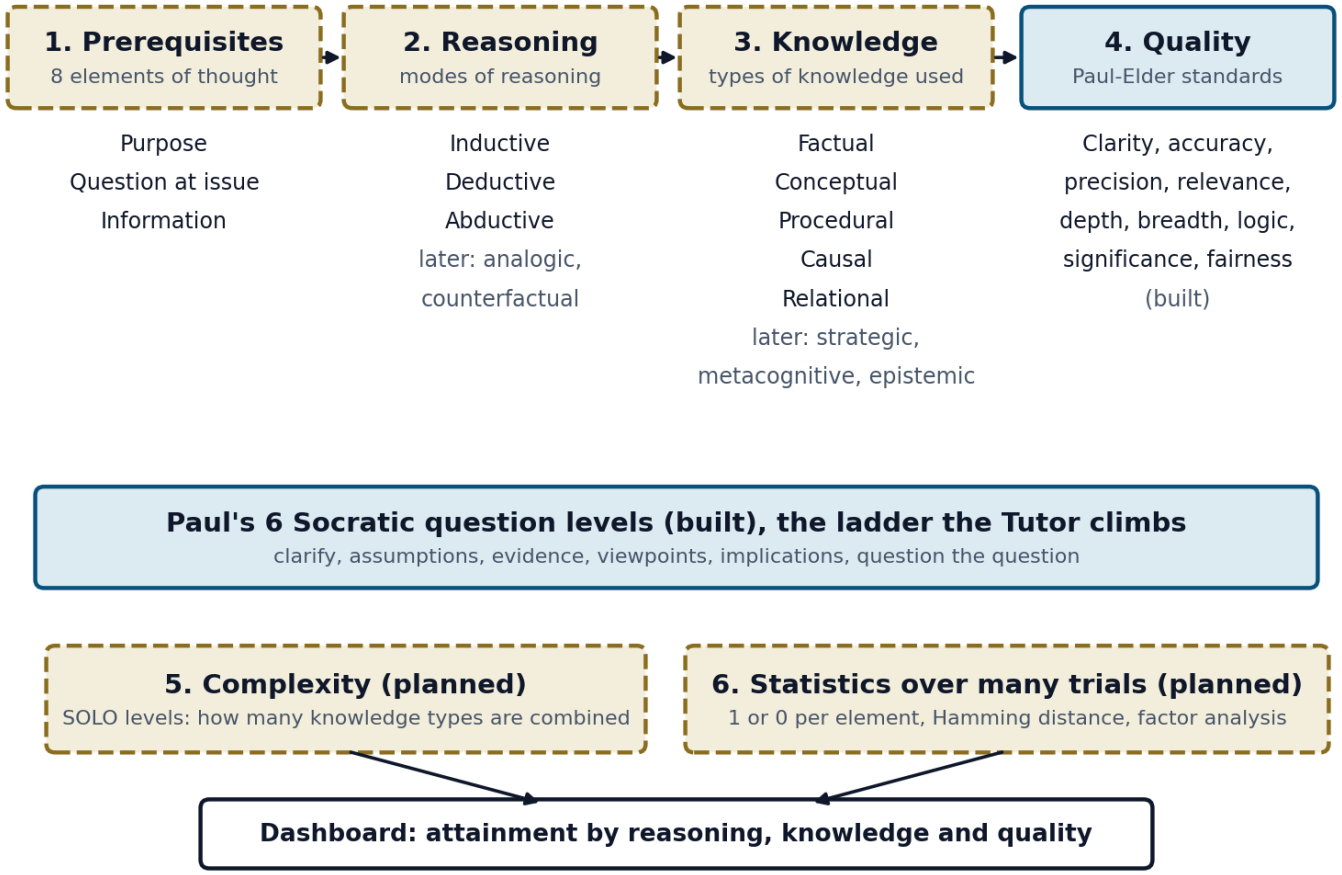


2. Feedback, one question, a hidden hint, and a simulation result

3, 6 or no-limit rounds | **Round and level shown** | **4 question formats** | **"I'm not sure" button** | **Status panel** | **Tutorial**

PROPOSED METHODOLOGY

Next Stage (Goal 1): Reasoning and Knowledge Layer



- Knowledge layer: each case note tagged with its type of knowledge; an agent drafts a knowledge graph
- Reasoning agent: 1 or 0 for inductive, deductive and abductive reasoning and each knowledge type, with evidence
- Question bank: several questions per skill, so there are enough trials
- Statistics: share of 1s and Hamming distance per learner [14]; factor analysis across learners
- Explainer agent: a plain scenario on why a skill matters

Example: "All the regional banks fell, not just CIMB" is inductive reasoning on relational knowledge.

PROPOSED METHODOLOGY

Evaluation Plan

1 Requirement check

Run the same scenarios on the baseline Gem tutors and on Socratic AI, and mark R1 to R9 as met or not.

2 Assessor vs human markers

Mark 61 test answers by hand with a marking guide. Compare using percent agreement and Cohen's kappa per standard (parent grant target 0.8).

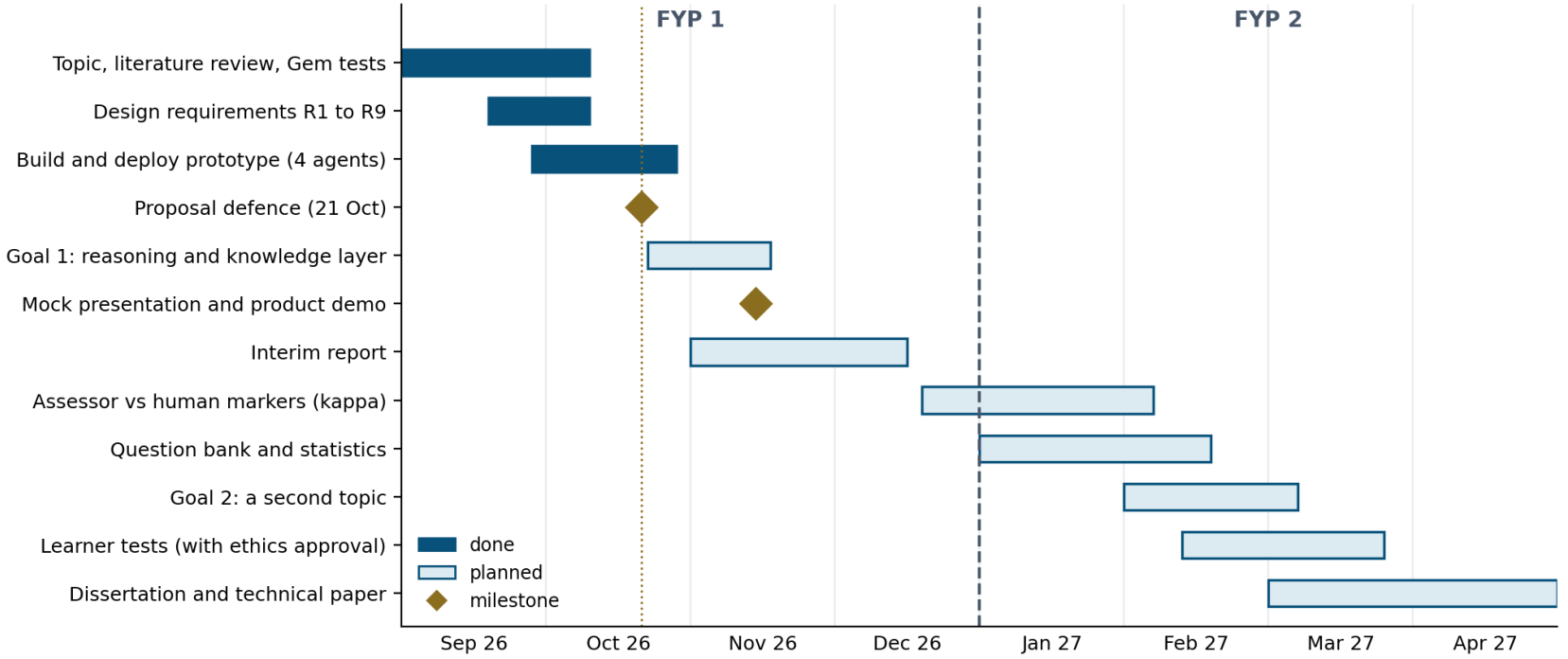
3 With and without agents

Switch the Planner, simulations or Checker off, and compare the same sessions.

4 Learner experience

A short questionnaire with Dr Ho's students, after ethics approval.

GANTT CHART & MILESTONES



Key Milestones

21 Oct 2026
Proposal defence

Mid Nov 2026
Goal 1 done, mock presentation

End of FYP 1
Interim report

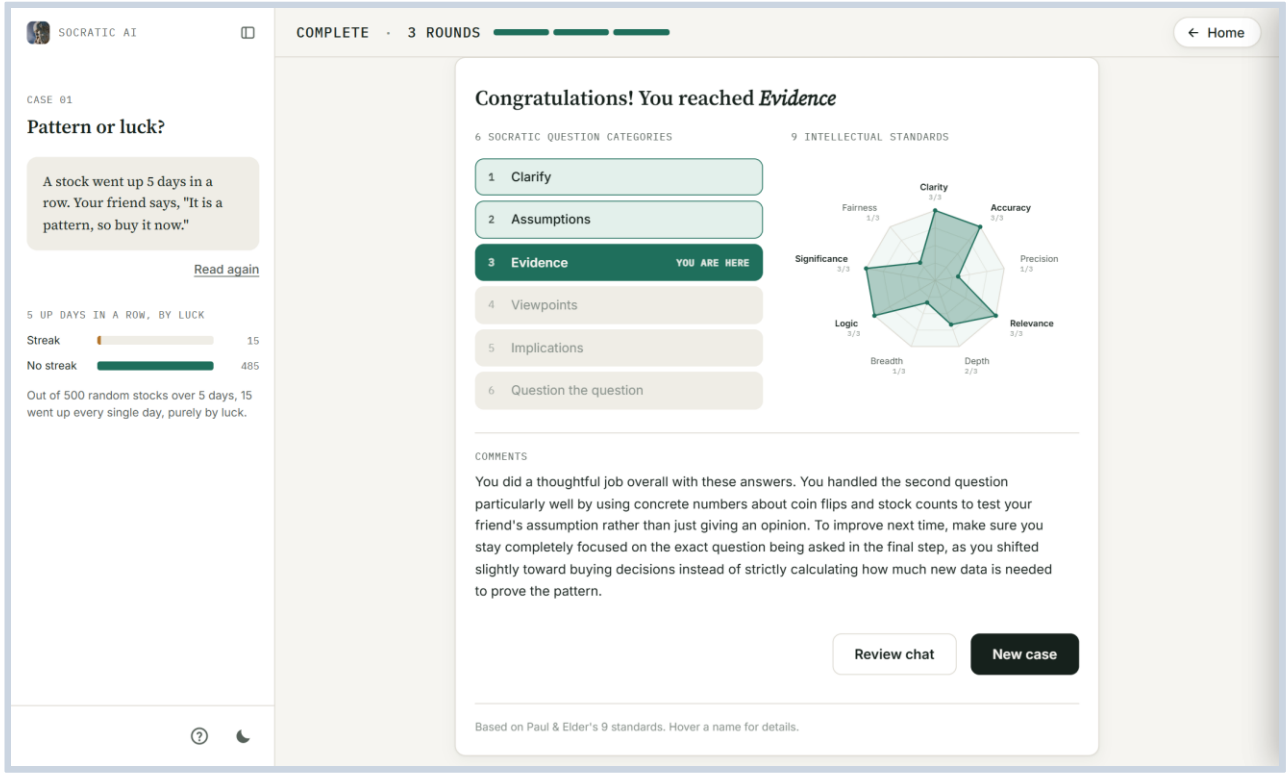
FYP 2
Assessor validation, learner tests, second topic

End of FYP 2
Dissertation and technical paper

FYP 2 dates are a first plan, to be confirmed with the supervisor.

PRELIMINARY RESULTS

A Real 3-Round Session (Gemini)



End profile: level reached, the 9 standards chart and the Coach comment

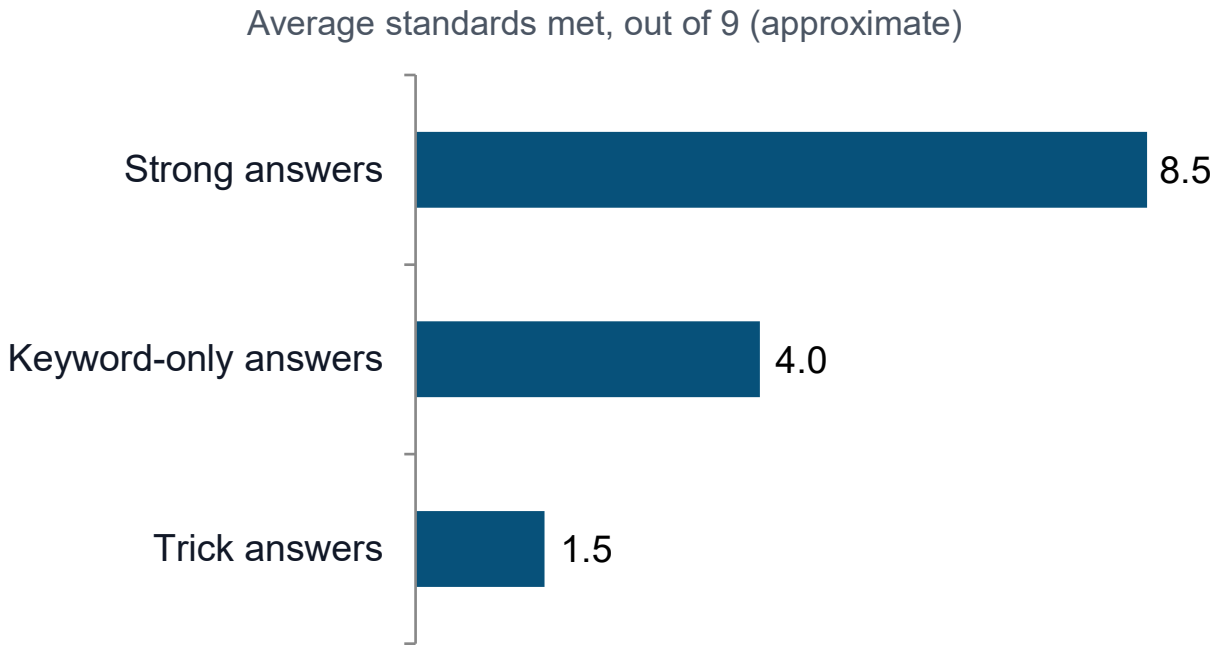
0 times the answer was given away, even when asked to ignore the rules

5 of 5 problems found in testing were fixed

~24 AI calls per session, inside the free tier

PRELIMINARY RESULTS

Assessor on a 61-Answer Test Set



37 real answers from test sessions and 24 constructed answers. Not yet compared with human markers, so this is not a validation yet.

Problems found and fixed

- The welcome said there was no round limit: now written by code
- The same question asked again in new words: the Checker now compares questions
- The Planner's simulation was ignored: the rules now run it
- Levels rose too fast: now 2 good answers in a row, or 1 strong answer
- The Assessor was generous: strict rules per standard; standards met fell from 24 to 16 on 12 real answers

Next: keyword-only answers still score about 4 of 9, so the Assessor prompt is tightened again.

CONCLUSIONS

Existing Socratic AI tutors have no question ladder and no real assessment. Socratic AI is a working multi-agent tutor that follows Paul's six levels, marks every answer against the nine standards, and never gives the answer away. Next, it will show which type of reasoning and knowledge a learner used, and be checked against human markers.

Take-home message

This is a reasoning trainer, not an answer machine, built in a new way: many prompted agents working together, guarded by code.

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